

Motion in a straight line

Kinematics describes motion without asking what causes it.

For constant acceleration there are three equations. Learn them as a set.

Key equations

$$v = u + at$$

$$s = ut + (1/2)at^2$$

$$v^2 = u^2 + 2as$$

Worked example

Worked example. A ball is thrown upward at 20 m/s. How high does it go? Take $g = 10 \text{ m/s}^2$.

At the top $v = 0$, so use the third equation with $a = -g$:

$$0 = 20^2 - 2(10)h$$

$$h = 400 / 20 = 20 \text{ m}$$

Exam tip: write the equation on its own line before you substitute.

Force, mass and acceleration

The net force on a body equals its mass times its acceleration.

A scale in a lift reads the normal force N , not your weight mg .

Key equations

$$F = ma$$

Apparent weight: $N - mg = ma$ for upward acceleration.

Downward acceleration: $mg - N = ma$

Worked example

Worked example. A 60 kg person stands in a lift accelerating upward at 2 m/s^2 . Take $g = 10 \text{ m/s}^2$.

$N - mg = ma$, so $N = m(g + a)$

$$N = 60(10 + 2) = 720 \text{ N}$$

Exam tip: decide the direction of the acceleration first, then write the net force.

Work, energy and the angle

Only the part of the force along the displacement does work.

Kinetic energy is the energy of motion.

Key equations

Work done by a force at an angle: $W = F s \cos(\theta)$

$E = \frac{1}{2} m v^2$

Worked example

Worked example. A force of 10 N moves a body 5 m at 60 degrees to the displacement.

$W = F s \cos(\theta) = 10 \times 5 \times 0.5$

$W = 25 \text{ J}$

Exam tip: state the formula, use $\cos(\theta)$, and finish with the unit, J.